

Embedded Systems Homework #2

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September 2023

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1 Directories and its information

1 `/etc`

- It contains configuration files and system-wide settings that are used by various programs and services on the system.

1 `/boot`

- It is containing files necessary for the system's initial boot process.

1 `/bin`

- This directory is one of the fundamental directories in a Unix-like file system, and it typically contains essential binary executable files necessary for the system's basic operations and recovery in single-user mode.

1 `/sbin`

- The directory is another important directory in Unix-like operating systems, and it contains system binaries (executables) that are used for system administration and maintenance tasks.

1 `/usr/bin`

- The `/usr/bin` directory is a crucial part of a Unix-like file system, and it contains user-level binary executables, commands, and programs that are available to all users of the system.

1 `/usr/sbin`

- The `/usr/sbin` directory is a significant part of Unix-like file systems, and it contains system administration binaries and commands that are intended for system management tasks performed after the system has fully booted.

1 `/dev`

- The `/dev` directory in Unix-like operating systems is a special directory that serves as the interface to device files. Device files represent physical and virtual devices on the system and provide a way for programs to interact with hardware components and devices.

1

`/var/log`

- The `/var/log` directory is a directory in Unix-like operating systems that is used to store log files generated by various system services, applications, and the system itself. Log files record important information about system events, errors, and activities, and they are essential for system administration, troubleshooting, and monitoring.

1

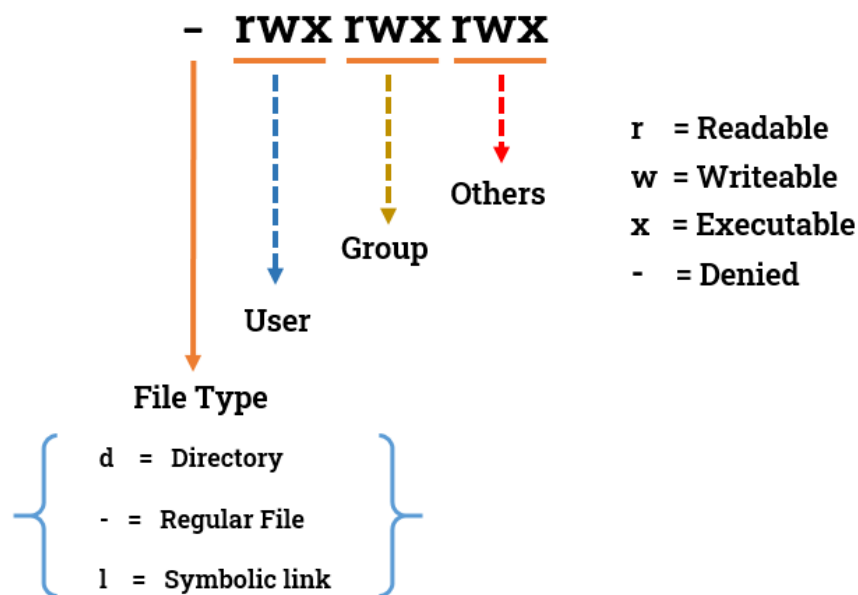
`/run`

- The `/run` directory is a relatively recent addition to Unix-like operating systems, and it serves as a location for storing runtime data and temporary files that are needed by the system during its operation.

2 Example queries

1. Traditional permissions

```
1  ls -l /usr/bin/passwd
2
3  Example output :
4
5  -rw-r--r-- 1 user1 users 12345 Sep 26 15:30
   myfile.txt
6  drwxr-xr-x 2 user2 users 4096 Sep 26 15:29
   mydir
```



Source: <https://linuxopsys.com/topics/linux-file-permissions>

2. File type

```
1  file -l /usr/bin/passwd
2
3  Example output:
4
5  myfile.txt: ASCII text
```

3. Hidden attributes of the file

```
1      ls -a -l /usr/bin/passwd
2
3      Example output :
4
5      total 24
6      drwxr-xr-x  4 user1 users 4096 Sep 26 15:30 .
7      drwxr-xr-x 12 user1 users 4096 Sep 26 15:29 ..
8      -rw-r--r--  1 user1 users 12345 Sep 26 15:30
9      myfile.txt
10     drwxr-xr-x  2 user1 users 4096 Sep 26 15:31
11     mydir
12     -rw-r--r--  1 user1 users   678 Sep 26 15:32 .
13     hiddenfile
```

The hiddenfile has a ” . ” in front of it

3 Firewall

Write a firewall file with this rules

1. Rule zero : Clear all existing rules (iptables -F...)
2. Default policy: Except for the custom chain INPUT which is set to DROP, the others are set to ACCEPT by default.
3. Trust this machine: Since lo is very important to this machine, lo must be set as a trusted device.
4. Response packet: Allow the machine to actively request and respond to packets that can enter the machine (ESTABLISHED, RELATED)
5. Trusted User: This is optional if you want to make your host resources available to zone network sources

```
1      #RULE ZERO
2      sudo iptables -F
3
4      #DEFAULT POLICY
5      sudo iptables -P INPUT DROP
6
7      #TRUST THIS MACHINE
8      sudo iptables -A INPUT -i lo -j ACCEPT
9
10     #RESPONSE PACKET
11     sudo iptables -A INPUT -m conntrack --ctstate
12         ESTABLISHED,RELATED -j ACCEPT
13
14     #TRUSTED USERS
15     sudo iptables -A INPUT -s 192.168.1.100 -p tcp
16         --dport 80 -j ACCEPT
17
18     #SAVE THE PROCESS
19     sudo invoke-rc.d iptables-persistent save
```